DAILY PRACTICE CALENDAR

Geometry Competition Math (Grade 8) • 2027–2028

34 weeks • Monday–Friday • 10 problems a day (about 30–40 minutes)

Experience first. Symbols second. Understanding always.

How to Use This Calendar

- **Mon–Thu:** 10 problems on the day's skill: **8 new + 2 review** from earlier weeks. Start with the **Try** problem as the warm-up.
- **Friday:** Weekly Quiz of 10 mixed problems (8 this week + 2 review). Copy every missed problem into the **error log** and redo it next Monday.
- **Holidays and breaks** are marked. A holiday's skill moves to the weekend; break weeks are not counted in the 34 weeks.
- This course follows Algebra 1 Competition Math: it deepens contest geometry for **AMC 8 and MATHCOUNTS** and prepares for **AMC 10**. Answers to every **Try** problem are on the last page.

Year at a Glance

Wk	Dates	Topic
1	Aug 16–Aug 20	Problem-Solving Toolkit for Geometry
2	Aug 23–Aug 27	Angles & Parallel Lines
3	Aug 30–Sep 3	Angles in Triangles
4	Sep 6–Sep 10	Polygon Angles
5	Sep 13–Sep 17	Triangle Inequality & Triangle Types
6	Sep 20–Sep 24	Congruent & Isosceles Triangles
7	Sep 27–Oct 1	The Pythagorean Theorem
8	Oct 4–Oct 8	Special Right Triangles
9	Oct 11–Oct 15	Similar Triangles
10	Oct 18–Oct 22	Review & Mini-Contest 1
11	Oct 25–Oct 29	Area of Triangles & Quadrilaterals
12	Nov 1–Nov 5	Area Ratios
13	Nov 8–Nov 12	Triangle Area Tricks
14	Nov 15–Nov 19	Coordinate Geometry
	Nov 22–Nov 26	*Thanksgiving Break*
15	Nov 29–Dec 3	Pick's Theorem & Lattice Geometry
16	Dec 6–Dec 10	Circle Basics
17	Dec 13–Dec 17	Inscribed Angles
	Dec 20–Dec 24	*Winter Break*
	Dec 27–Dec 31	*Winter Break*
18	Jan 3–Jan 7	Tangents & Chords
19	Jan 10–Jan 14	Power of a Point
20	Jan 17–Jan 21	Mid-Year Mock Contest & Review
21	Jan 24–Jan 28	Prisms & Cylinders
22	Jan 31–Feb 4	Pyramids, Cones & Spheres
23	Feb 7–Feb 11	Surface Area & Nets
24	Feb 14–Feb 18	Distances in 3D
25	Feb 21–Feb 25	Transformations & Symmetry
26	Feb 28–Mar 3	Scaling Area & Volume
27	Mar 6–Mar 10	Counting Figures
28	Mar 13–Mar 17	Lattice Paths
29	Mar 20–Mar 24	Geometric Probability
	Mar 27–Mar 31	*Spring Break*
30	Apr 3–Apr 7	Diagonals & Euler's Formula
31	Apr 10–Apr 14	Geometric Optimization
32	Apr 17–Apr 21	Contest Strategy: AMC 8 & AMC 10 Geometry
33	Apr 24–Apr 28	Full Mock Contest & Review
34	May 1–May 5	Final Contest & Celebration

Units 1 & 6 · Unit 2 · Unit 3 · Unit 4 · Unit 5 · Wrap-Up · Break

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WEEK 1 • Aug 16–Aug 20, 2027 — Unit 1: Angles, Lines & Polygons

Problem-Solving Toolkit for Geometry — **Core:** Draw it big, label everything, and add a helpful line.

Mon Aug 16
Draw & label
TRY A square has perimeter 36. Find its area.

Tue Aug 17
Add a helpful line
TRY A rectangle is 8 by 6. How long is its diagonal?

Wed Aug 18
Split into pieces
TRY Four 3 × 3 squares are cut from the corners of a 10 × 10 square. What area is left?

Thu Aug 19
Work backward from area
TRY A rectangle has area 48 and length 8. Find its perimeter.

Fri Aug 20
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 2 • Aug 23–Aug 27, 2027 — Unit 1: Angles, Lines & Polygons

Angles & Parallel Lines — **Core:** Vertical angles are equal; parallel lines copy angles.

Mon Aug 23
Angle pairs
TRY Two supplementary angles differ by $40°$. Find the larger angle.

Tue Aug 24
Vertical angles
TRY Vertical angles measure $(5x - 20)°$ and $(3x + 30)°$. Find x.

Wed Aug 25
Parallel lines
TRY Parallel lines are cut by a transversal. One same-side interior angle is 3 times the other. Find the smaller angle.

Thu Aug 26
Clock angles
TRY What is the smaller angle between the clock hands at 3:30?

Fri Aug 27
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 3 • Aug 30–Sep 3, 2027 — Unit 1: Angles, Lines & Polygons

Angles in Triangles — **Core:** Triangle angles add to $180°$; an exterior angle equals the two remote interior angles.

Mon Aug 30
Angle sum
TRY A triangle has angles x, $2x$ and $3x$. Find the largest angle.

Tue Aug 31
Exterior angles
TRY An exterior angle of a triangle is $130°$, and one remote interior angle is $55°$. Find the other.

Wed Sep 1
Isosceles triangles
TRY An isosceles triangle has vertex angle $36°$. Find each base angle.

Thu Sep 2
Angle chasing
TRY In $\triangle ABC$, $AB = AC$ and $\angle A = 40°$. The bisector of $\angle B$ meets AC at D. Find $\angle BDC$.

Fri Sep 3
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 4 • Sep 6–Sep 10, 2027 — Unit 1: Angles, Lines & Polygons

Polygon Angles — **Core:** Interior angles of an n-gon add to $(n - 2) \times 180°$; exterior angles add to $360°$.

Mon Sep 6
HOLIDAY: Labor Day
No class. Do this skill's 10 problems on the weekend:
Interior angle sum
TRY Find the sum of the interior angles of an octagon.

Tue Sep 7
Regular polygons
TRY Each exterior angle of a regular polygon is $24°$. How many sides does it have?

Wed Sep 8
Missing angle
TRY A pentagon has angles $100°$, $110°$, $120°$ and $95°$. Find the fifth angle.

Thu Sep 9
Star angles
TRY Find the sum of the 5 tip angles of a five-pointed star.

Fri Sep 10
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 5 • Sep 13–Sep 17, 2027 — Unit 2: Triangles

Triangle Inequality & Triangle Types — **Core:** Any two sides must add to more than the third.

Mon Sep 13
Triangle inequality
TRY A triangle has sides 7 and 12. How many integer lengths are possible for the third side?

Tue Sep 14
Biggest side, biggest angle
TRY A triangle has sides 5, 7 and 9. Which side is opposite the largest angle?

Wed Sep 15
Acute, right or obtuse?
TRY Is a triangle with sides 6, 8 and 11 acute, right or obtuse?

Thu Sep 16
Counting triangles
TRY How many different triangles with integer sides have perimeter 12?

Fri Sep 17
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 6 • Sep 20–Sep 24, 2027 — Unit 2: Triangles

Congruent & Isosceles Triangles — **Core:** Equal sides face equal angles; SSS, SAS and ASA prove congruence.

Mon Sep 20
Congruence shortcuts
TRY Which of SSS, SAS, ASA and SSA does not always prove two triangles congruent?

Tue Sep 21
Isosceles algebra
TRY A triangle has sides $2x+3$, $3x-1$ and 15, and the first two are equal. Find its perimeter.

Wed Sep 22
Equilateral triangles
TRY An equilateral triangle has perimeter 27. Find its height.

Thu Sep 23
Perpendicular bisectors
TRY P is on the perpendicular bisector of AB, $PA = 13$ and $AB = 10$. How far is P from line AB?

Fri Sep 24
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 7 • Sep 27–Oct 1, 2027 — Unit 2: Triangles

The Pythagorean Theorem — **Core:** Know the triples: 3-4-5, 5-12-13, 8-15-17, 7-24-25.

Mon Sep 27
Triples
TRY A right triangle has legs 15 and 8. Find the hypotenuse.

Tue Sep 28
Ladders
TRY A 25-ft ladder leans on a wall with its foot 7 ft from the wall. How high does it reach?

Wed Sep 29
Rectangles & diagonals
TRY A rectangle has diagonal 13 and one side 5. Find its area.

Thu Sep 30
Rhombus
TRY A rhombus has diagonals 10 and 24. Find its side length.

Fri Oct 1
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 8 • Oct 4–Oct 8, 2027 — Unit 2: Triangles

Special Right Triangles — **Core:** 45-45-90 sides are $1:1:\sqrt{2}$; 30-60-90 sides are $1:\sqrt{3}:2$.

Mon Oct 4
45-45-90 triangles
TRY A square has diagonal $8\sqrt{2}$. Find its side.

Tue Oct 5
30-60-90 triangles
TRY A 30-60-90 triangle has shorter leg 7. Find the hypotenuse and the longer leg.

Wed Oct 6
Equilateral area
TRY Find the area of an equilateral triangle with side 6.

Thu Oct 7
Regular hexagons
TRY Find the area of a regular hexagon with side 4.

Fri Oct 8
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 9 • Oct 11–Oct 15, 2027 — Unit 2: Triangles

Similar Triangles **Core:** Similar triangles have equal angles and proportional sides.

Mon Oct 11
Scale factor
TRY A triangle with sides 4, 6, 8 is similar to one whose shortest side is 10. Find its longest side.

Tue Oct 12
A parallel line
TRY In $\triangle ABC$, $DE \parallel BC$ with D on AB and E on AC. $AD = 3$, $DB = 6$, $BC = 12$. Find DE.

Wed Oct 13
Shadows & heights
TRY A 1.5 m post casts a 2 m shadow. A building casts a 24 m shadow. How tall is the building?

Thu Oct 14
Area ratios
TRY Two similar triangles have sides in the ratio $2:5$. What is the ratio of their areas?

Fri Oct 15
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 10 • Oct 18–Oct 22, 2027 — Unit 2: Triangles

Review & Mini-Contest 1 **Core:** Pull Units 1–2 together and practice contest habits.

Mon Oct 18
Review: polygon angles
TRY Each interior angle of a regular polygon is 140°. How many sides?

Tue Oct 19
Review: right triangles
TRY An isosceles right triangle has hypotenuse 10. Find its area.

Wed Oct 20
Review: similarity
TRY A 3×5 rectangle is similar to a rectangle with short side 9. Find its long side.

Thu Oct 21
Timed set: 10 problems in 20 min
TRY Find the area of a triangle with sides 9, 12 and 15.

Fri Oct 22
Mini-Contest 1
Units 1–2, timed. Afterward, start an error log of every missed problem.

WEEK 11 • Oct 25–Oct 29, 2027 — Unit 3: Area & Coordinate Geometry

Area of Triangles & Quadrilaterals **Core:** Find a height you can compute, often by drawing an altitude.

Mon Oct 25
Isosceles triangles
TRY Find the area of a triangle with sides 13, 13 and 10.

Tue Oct 26
Trapezoids
TRY An isosceles trapezoid has bases 10 and 22 and legs 10. Find its area.

Wed Oct 27
Kites & rhombuses
TRY A kite has diagonals 12 and 9. Find its area.

Thu Oct 28
Midpoint squares
TRY Joining the midpoints of the sides of a square with side 10 makes a smaller square. Find its area.

Fri Oct 29
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 12 • Nov 1–Nov 5, 2027 — Unit 3: Area & Coordinate Geometry

Area Ratios **Core:** Triangles with the same height have areas in the same ratio as their bases.

Mon Nov 1
Same height
TRY In $\triangle ABC$, D is on BC with $BD : DC = 2 : 3$. The area of $\triangle ABC$ is 40. Find the area of $\triangle ABD$.

Tue Nov 2
Medians
TRY The three medians cut a triangle of area 36 into 6 small triangles. What is the area of each?

Wed Nov 3
Midpoint triangle
TRY Joining the midpoints of a triangle of area 48 forms a smaller triangle. Find its area.

Thu Nov 4
Parallelograms
TRY $ABCD$ is a parallelogram of area 60, and P is any point on CD. Find the area of $\triangle ABP$.

Fri Nov 5
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 13 • Nov 8–Nov 12, 2027 — Unit 3: Area & Coordinate Geometry

Triangle Area Tricks

Core: Pick the formula that fits: $\frac{1}{2}bh$, Heron's formula, or area $= r \cdot s$.

Mon Nov 8
Heron's formula
TRY Find the area of a triangle with sides 5, 5 and 6.

Tue Nov 9
The 13-14-15 triangle
TRY Find the area of a triangle with sides 13, 14 and 15.

Wed Nov 10
Inradius
TRY Using area $= r \cdot s$ (s = half the perimeter), find the inradius of the 13-14-15 triangle.

Thu Nov 11
HOLIDAY: Veterans Day
No class. Do this skill's 10 problems on the weekend:
Altitudes from area
TRY In the 13-14-15 triangle, find the altitude to the side of length 14.

Fri Nov 12
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 14 • Nov 15–Nov 19, 2027 — Unit 3: Area & Coordinate Geometry

Coordinate Geometry

Core: Distance, midpoint and slope — plus the shoelace formula for area.

Mon Nov 15
Distance formula
TRY Find the distance between $(-1, 2)$ and $(5, 10)$.

Tue Nov 16
Shoelace formula
TRY Find the area of the triangle with vertices $(0, 0)$, $(4, 1)$ and $(2, 5)$.

Wed Nov 17
Regions bounded by lines
TRY Find the area of the triangle bounded by $y = 2x$, $y = -x + 9$ and the x-axis.

Thu Nov 18
Lattice points on circles
TRY How many points with integer coordinates lie on the circle $x^2 + y^2 = 25$?

Fri Nov 19
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Thanksgiving Break • Nov 22–Nov 26, 2027 • no class (not counted in the 34 weeks)

WEEK 15 • Nov 29–Dec 3, 2027 — Unit 3: Area & Coordinate Geometry

Pick's Theorem & Lattice Geometry

Core: For a lattice polygon: area $= I + \frac{B}{2} - 1$.

Mon Nov 29
Pick's theorem
TRY A lattice polygon has 10 interior points and 8 boundary points. Find its area.

Tue Nov 30
Points on a segment
TRY How many lattice points lie on the segment from $(0, 0)$ to $(15, 9)$, endpoints included?

Wed Dec 1
Interior points
TRY How many lattice points lie strictly inside the triangle with vertices $(0, 0)$, $(8, 0)$, $(0, 6)$?

Thu Dec 2
Inside a rectangle
TRY How many lattice points lie strictly inside the rectangle with corners $(0, 0)$ and $(7, 5)$?

Fri Dec 3
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 16 • Dec 6–Dec 10, 2027 — Unit 4: Circles

Circle Basics

Core: Arcs and sectors are fractions of the whole circle.

Mon Dec 6
Arc length
TRY A circle has radius 9. Find the length of an arc with central angle $80°$.

Tue Dec 7
Sector area
TRY A circle has radius 10. Find the area of a sector with central angle $72°$.

Wed Dec 8
Chords
TRY A chord of length 16 is in a circle of radius 10. How far is it from the center?

Thu Dec 9
Nested figures
TRY A circle is inscribed in a square of side 12, and a square is inscribed in that circle. Find the small square's area.

Fri Dec 10
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 17 • Dec 13–Dec 17, 2027 — Unit 4: Circles

Inscribed Angles **Core:** An inscribed angle is half of the arc it cuts off.

Mon Dec 13
Inscribed angles
TRY An inscribed angle cuts off an arc of 110°. Find the angle.

Tue Dec 14
Angles in a semicircle
TRY AB is a diameter and C is on the circle. $AC = 6$ and $BC = 8$. Find the radius.

Wed Dec 15
Cyclic quadrilaterals
TRY In cyclic quadrilateral $ABCD$, $\angle A = 75^\circ$. Find $\angle C$.

Thu Dec 16
Central vs. inscribed
TRY A, B, C are on a circle with center O, and $\angle ACB = 35^\circ$. Find $\angle AOB$.

Fri Dec 17
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Winter Break • Dec 20–Dec 24, 2027 • no class (not counted in the 34 weeks)

Winter Break • Dec 27–Dec 31, 2027 • no class (not counted in the 34 weeks)

WEEK 18 • Jan 3–Jan 7, 2028 — Unit 4: Circles

Tangents & Chords **Core:** A tangent is perpendicular to the radius; two tangents from one point are equal.

Mon Jan 3
Tangent length
TRY Point P is 13 from the center of a circle of radius 5. Find the length of a tangent from P.

Tue Jan 4
Equal tangents
TRY A circle is inscribed in quadrilateral $ABCD$ with $AB = 8$, $BC = 11$, $CD = 13$. Find DA.

Wed Jan 5
Inradius of a right triangle
TRY Find the radius of the circle inscribed in a 6-8-10 triangle.

Thu Jan 6
Chord angles
TRY Two chords cross inside a circle, cutting off arcs of 70° and 110°. Find the angle between them.

Fri Jan 7
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 19 • Jan 10–Jan 14, 2028 — Unit 4: Circles

Power of a Point **Core:** For every line through P, $PA \cdot PB$ is the same.

Mon Jan 10
Crossing chords
TRY Chords AB and CD meet at P with $AP = 4$, $PB = 9$ and $CP = 6$. Find PD.

Tue Jan 11
Two secants
TRY From an outside point P: $PA = 4$, $PB = 12$ on one secant and $PC = 6$ on another. Find PD.

Wed Jan 12
Tangent & secant
TRY Tangent $PT = 6$. A secant from P has $PA = 4$. Find PB.

Thu Jan 13
Power inside
TRY P is 6 from the center of a circle of radius 10. A chord through P is split into two parts. Find their product.

Fri Jan 14
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 20 • Jan 17–Jan 21, 2028 — Unit 4: Circles

Mid-Year Mock Contest & Review **Core:** Review Units 1–4 and practice pacing a full contest.

Mon Jan 17
HOLIDAY: MLK Jr. Day
No class. Do this skill's 10 problems on the weekend:
Review: triangles
TRY Find the area of a triangle with sides 7, 24 and 25.

Tue Jan 18
Review: area ratios
TRY Two similar triangles have perimeters 12 and 18. The smaller has area 20. Find the larger area.

Wed Jan 19
Review: circles
TRY An inscribed angle is 40°. What arc does it cut off?

Thu Jan 20
Pacing: skip & return
TRY 15-minute set. Warm-up: find the circumference of a circle with area 49π.

Fri Jan 21
Mid-Year Mock Contest
Full-length, Units 1–4. Then sort errors by topic in the error log.

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WEEK 21 • Jan 24–Jan 28, 2028 — Unit 5: Solids & Transformations

Prisms & Cylinders **Core:** Volume = base area × height.

Mon Jan 24
Boxes from face areas
TRY A box has faces of area 12, 15 and 20. Find its volume.

Tue Jan 25
Cylinders
TRY A cylinder has volume 72π and radius 3. Find its height.

Wed Jan 26
Triangular prisms
TRY A prism has a right-triangle base with legs 3 and 4, and height 10. Find its volume.

Thu Jan 27
Water displacement
TRY A cube of edge 4 is dropped into an 8 × 8 tank of water. How much does the water rise?

Fri Jan 28
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 22 • Jan 31–Feb 4, 2028 — Unit 5: Solids & Transformations

Pyramids, Cones & Spheres **Core:** A pointed solid is $\frac{1}{3}$ of its prism; a sphere has $V = \frac{4}{3}\pi r^3$.

Mon Jan 31
Pyramids
TRY A square pyramid has base edge 6 and height 10. Find its volume.

Tue Feb 1
Cones
TRY A cone has radius 3 and height 4. Find its volume and slant height.

Wed Feb 2
Spheres
TRY Find the volume and surface area of a sphere with radius 3.

Thu Feb 3
Sphere in a cube
TRY Find the volume of the largest sphere that fits inside a cube of edge 10.

Fri Feb 4
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 23 • Feb 7–Feb 11, 2028 — Unit 5: Solids & Transformations

Surface Area & Nets **Core:** Unfold the solid: surface area is the area of its net.

Mon Feb 7
Cubes
TRY A cube has surface area 150. Find its volume.

Tue Feb 8
Cylinders
TRY Find the total surface area of a cylinder with radius 2 and height 5.

Wed Feb 9
Painted cubes
TRY A 4 × 4 × 4 cube is painted, then cut into unit cubes. How many have exactly two painted faces?

Thu Feb 10
Cones
TRY Find the curved (lateral) area of a cone with radius 5 and slant height 13.

Fri Feb 11
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 24 • Feb 14–Feb 18, 2028 — Unit 5: Solids & Transformations

Distances in 3D **Core:** Use the Pythagorean theorem twice for a space diagonal.

Mon Feb 14
Space diagonals
TRY Find the space diagonal of a 3 × 4 × 12 box.

Tue Feb 15
Cube diagonals
TRY A cube has edge 5. Find its space diagonal.

Wed Feb 16
The ant on a cube
TRY An ant crawls on the surface of a unit cube between opposite corners. Find the shortest path.

Thu Feb 17
Face diagonals
TRY In a cube of edge 2, three face diagonals form an equilateral triangle. Find its area.

Fri Feb 18
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 25 • Feb 21–Feb 25, 2028 — Unit 5: Solids & Transformations

Transformations & Symmetry

Core: Slide, flip or turn — and use reflections to find shortest paths.

Mon Feb 21

HOLIDAY: Presidents' Day

No class. Do this skill's 10 problems on the weekend:

Reflections

TRY Reflect $(3, -5)$ over the line $y = x$.

Tue Feb 22

Rotations

TRY Rotate $(4, 1)$ by $180°$ about the point $(2, 2)$.

Wed Feb 23

Lines of symmetry

TRY How many lines of symmetry does a regular hexagon have?

Thu Feb 24

Shortest path by reflection

TRY $A(0, 3)$ and $B(6, 5)$ are above a river along the x-axis. Find the shortest path from A to the river to B.

Fri Feb 25

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 26 • Feb 28–Mar 3, 2028 — Unit 5: Solids & Transformations

Scaling Area & Volume

Core: Scale lengths by k: area scales by k^2, volume by k^3.

Mon Feb 28

Scaling area

TRY A photo is enlarged so each side is 3 times as long. Its area is multiplied by what?

Tue Feb 29

Scaling volume

TRY Two similar cans have heights 6 and 9. The small one holds 16 oz. How much does the large one hold?

Wed Mar 1

Cutting a cone

TRY A cone is cut parallel to its base halfway up. What fraction of the volume is the small top cone?

Thu Mar 2

Similar solids

TRY Two similar solids have surface areas in the ratio 4 : 9. Find the ratio of their volumes.

Fri Mar 3

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 27 • Mar 6–Mar 10, 2028 — Unit 6: Counting & Probability in Geometry

Counting Figures

Core: Count by size or by corners so nothing is missed.

Mon Mar 6

Rectangles in a grid

TRY How many rectangles are in a 3 × 3 grid of squares?

Tue Mar 7

Squares in a grid

TRY How many squares of all sizes are in a 5 × 5 grid?

Wed Mar 8

Points on a circle

TRY How many triangles have their vertices among 8 points on a circle?

Thu Mar 9

Points in a grid

TRY How many triangles have their vertices among the 9 points of a 3 × 3 square lattice?

Fri Mar 10

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 28 • Mar 13–Mar 17, 2028 — Unit 6: Counting & Probability in Geometry

Lattice Paths

Core: A path of rights and ups is a choice of which steps go right.

Mon Mar 13

Basic paths

TRY Moving only right or up one unit at a time, how many paths go from $(0, 0)$ to $(5, 3)$?

Tue Mar 14

Through a point

TRY How many such paths go from $(0, 0)$ to $(4, 4)$ through $(2, 2)$?

Wed Mar 15

Avoiding a point

TRY How many such paths go from $(0, 0)$ to $(4, 4)$ without passing through $(2, 2)$?

Thu Mar 16

Paths in 3D

TRY Moving one unit in the $+x$, $+y$ or $+z$ direction, how many paths go from $(0, 0, 0)$ to $(2, 2, 2)$?

Fri Mar 17

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 29 • Mar 20–Mar 24, 2028 — Unit 6: Counting & Probability in Geometry

Geometric Probability **Core:** Probability = favorable length, area or volume ÷ total.

Mon Mar 20
On a segment
TRY A point is picked on a segment of length 10. P(it is within 2 of an endpoint)?

Tue Mar 21
Dartboards
TRY A dartboard has circles of radius 2 and 6 with the same center. A dart lands at random on the board. P(inner circle)?

Wed Mar 22
In a square
TRY A point (x, y) is picked at random in the unit square. P($x > 2y$)?

Thu Mar 23
Meeting problems
TRY Two friends arrive at random times between 12:00 and 1:00, and each waits 15 minutes. P(they meet)?

Fri Mar 24
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Spring Break • Mar 27–Mar 31, 2028 • no class (not counted in the 34 weeks)

WEEK 30 • Apr 3–Apr 7, 2028 — Unit 6: Counting & Probability in Geometry

Diagonals & Euler's Formula **Core:** An n-gon has $\frac{n(n-3)}{2}$ diagonals; a polyhedron has $V - E + F = 2$.

Mon Apr 3
Counting diagonals
TRY How many diagonals does a 12-sided polygon have?

Tue Apr 4
Working backward
TRY A polygon has 35 diagonals. How many sides does it have?

Wed Apr 5
Euler's formula
TRY A polyhedron has 12 vertices and 30 edges. How many faces does it have?

Thu Apr 6
Counting edges
TRY A soccer ball is made of 12 pentagons and 20 hexagons. How many edges does it have?

Fri Apr 7
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 31 • Apr 10–Apr 14, 2028 — Unit 6: Counting & Probability in Geometry

Geometric Optimization **Core:** The best shape is often the most symmetric one.

Mon Apr 10
Fencing against a wall
TRY 60 m of fence makes three sides of a rectangle; a wall is the fourth. Find the largest possible area.

Tue Apr 11
Distance to a line
TRY Find the distance from the origin to the line $3x + 4y = 20$.

Wed Apr 12
Largest triangle
TRY A triangle has two sides of length 8 and 10. What is its largest possible area?

Thu Apr 13
Least surface area
TRY A box has volume 64. What is its smallest possible surface area?

Fri Apr 14
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 32 • Apr 17–Apr 21, 2028 — Contest Season Wrap-Up

Contest Strategy: AMC 8 & AMC 10 Geometry **Core:** Draw accurately, mark equal lengths, and estimate from the diagram.

Mon Apr 17
Draw to scale
TRY Square $ABCD$ has side 6, and E is the midpoint of AB. Find the area of $\triangle ECD$.

Tue Apr 18
Hidden right triangles
TRY A circle of radius 5 is centered at the origin. Find the length of its chord on the line $y = 3$.

Wed Apr 19
AMC 10 style
TRY A right triangle has perimeter 30 and hypotenuse 13. Find its area.

Thu Apr 20
Timed set: 10 in 20 min
TRY What is the smaller angle between the clock hands at 2:20?

Fri Apr 21
Weekly Quiz
10 mixed contest problems, timed (25 min).

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WEEK 33 • Apr 24–Apr 28, 2028 — Contest Season Wrap-Up

Full Mock Contest & Review **Core:** Simulate contest day and learn from every error.

Mon Apr 24
Review: triangles & circles
TRY An equilateral triangle is inscribed in a circle of radius 4. Find its side.

Tue Apr 25
Review: solids
TRY A cube has volume 27. Find its space diagonal.

Wed Apr 26
Review: counting
TRY How many triangles have their vertices among the vertices of a regular hexagon?

Thu Apr 27
Full Mock Contest
Full-length mixed contest under real time limits.

Fri Apr 28
Error-log review
Redo every missed problem from the mock; write the key idea for each.

WEEK 34 • May 1–May 5, 2028 — Contest Season Wrap-Up

Final Contest & Celebration **Core:** Show what you have learned and set next year's goals.

Mon May 1
Light review: favorite strategies
TRY Find each interior angle of a regular decagon.

Tue May 2
Mixed warm-up set
TRY Find the area of a circle with circumference 20π.

Wed May 3
Final Contest
Full-length contest covering the whole year.

Thu May 4
Contest review
Go over solutions together; students present one problem each.

Fri May 5
Celebration & reflection
Certificates, progress review, and goals for next year.

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ANSWER KEY

Answers to the Daily "Try" Problems

Week 1
Mon: 81
Tue: 10
Wed: 64
Thu: 28

Week 2
Mon: 110°
Tue: $x = 25$
Wed: 45°
Thu: 75°

Week 3
Mon: 90°
Tue: 75°
Wed: 72°
Thu: 75°

Week 4
Mon: 1080°
Tue: 15
Wed: 115°
Thu: 180°

Week 5
Mon: 13
Tue: The side of length 9
Wed: Obtuse ($36 + 64 < 121$)
Thu: 3

Week 6
Mon: SSA
Tue: 37
Wed: $\frac{9\sqrt{3}}{2}$
Thu: 12

Week 7
Mon: 17
Tue: 24 ft
Wed: 60
Thu: 13

Week 8
Mon: 8
Tue: 14 and $7\sqrt{3}$
Wed: $9\sqrt{3}$
Thu: $24\sqrt{3}$

Week 9
Mon: 20
Tue: 4
Wed: 18 m
Thu: 4 : 25

Week 10
Mon: 9
Tue: 25
Wed: 15
Thu: 54

Week 11
Mon: 60
Tue: 128
Wed: 54
Thu: 50

Week 12
Mon: 16
Tue: 6
Wed: 12
Thu: 30

Week 13
Mon: 12
Tue: 84
Wed: 4
Thu: 12

Week 14
Mon: 10
Tue: 9
Wed: 27
Thu: 12

Week 15
Mon: 13
Tue: 4
Wed: 17
Thu: 24

Week 16
Mon: 4π
Tue: 20π
Wed: 6
Thu: 72

Week 17
Mon: 55°
Tue: 5
Wed: 105°
Thu: 70°

Week 18
Mon: 12
Tue: 10
Wed: 2
Thu: 90°

Week 19
Mon: 6
Tue: 8
Wed: 9
Thu: 64

Week 20
Mon: 84
Tue: 45
Wed: 80°
Thu: 14π

Week 21
Mon: 60
Tue: 8
Wed: 60
Thu: 1

Week 22
Mon: 120
Tue: 12π; slant height 5
Wed: Both 36π
Thu: $\frac{500\pi}{3}$

Week 23
Mon: 125
Tue: 28π
Wed: 24
Thu: 65π

Week 24
Mon: 13
Tue: $5\sqrt{3}$
Wed: $\sqrt{5}$
Thu: $2\sqrt{3}$

Week 25
Mon: $(-5, 3)$
Tue: $(0, 3)$
Wed: 6
Thu: 10

Week 26
Mon: 9
Tue: 54 oz
Wed: $\frac{1}{8}$
Thu: 8 : 27

Week 27
Mon: 36
Tue: 55
Wed: 56
Thu: 76

Week 28
Mon: 56
Tue: 36
Wed: 34
Thu: 90

Week 29
Mon: $\frac{2}{5}$
Tue: $\frac{1}{9}$
Wed: $\frac{1}{4}$
Thu: $\frac{7}{16}$

Week 30
Mon: 54
Tue: 10
Wed: 20
Thu: 90

Week 31
Mon: 450 m^2
Tue: 4
Wed: 40
Thu: 96 (a $4 \times 4 \times 4$ cube)

Week 32
Mon: 18
Tue: 8
Wed: 30
Thu: 50°

Week 33
Mon: $4\sqrt{3}$
Tue: $3\sqrt{3}$
Wed: 20

Week 34
Mon: 144°
Tue: 100π

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